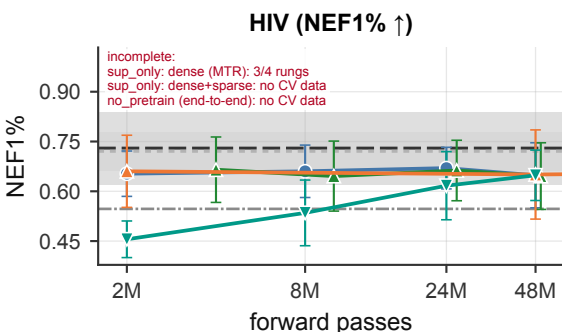
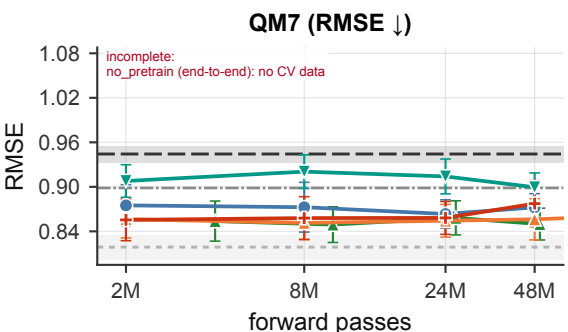
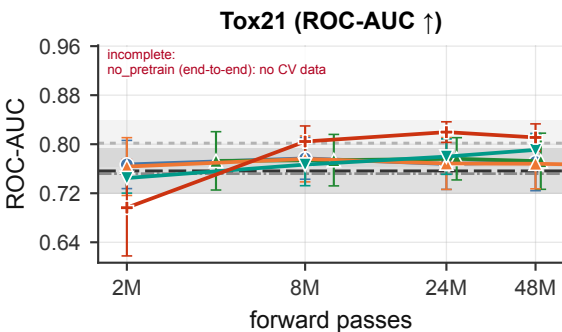
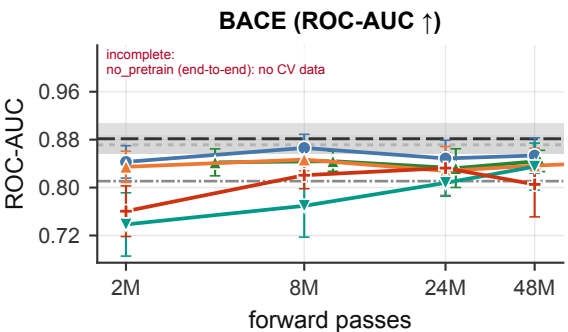
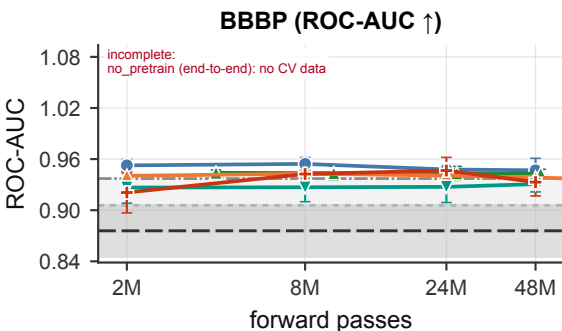
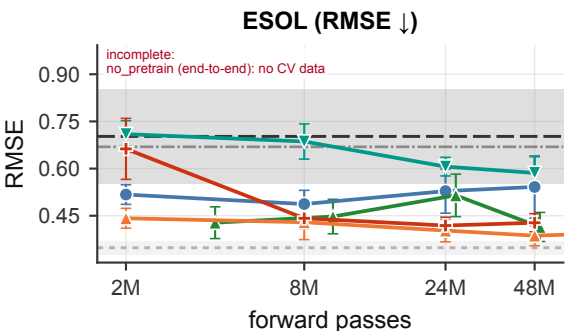


Fig A2.a - scaling of the primary regimes in forward passes

pooled 5-fold scaffold CV ONLY · every error bar and band = ± 1 sd across the 5 folds (no head-seed, no pretraining-seed spread), so every interval means the same thing · the 96M rung and unsup→sup's 48M rung are hold-out-only, so they are absent rather than interpolated · unsup→sup is at its TRUE total (MLM base + 2M-PP SFT)



- unsup_only (MLM)

★ unsup→sup: dense (MTR)

★ sup only: dense (MTR)

➡ sup_only: sparse all

✚ sup_only: dense+sparse

- Morgan+XGBoost (ECFP4)

- - - Morgan+desc+XGBoost

```
.... no_pretrain (end-to-end)
```

— no pretrain (floor)